

INTERESTS

Higher structures, derived geometry, lattice-based cryptography, homomorphic encryption

EDUCATION

Middle East Technical University
BSc in Mathematics, GPA: 4.00/4.00

Ankara, TR
August 28, 2022 – July 15, 2026

RESEARCH EXPERIENCE

Middle East Technical University
Higher Structures Group

Ankara, TR
March 30, 2026 – Current

- Investigation of shifted contact structures on derived stacks with Kadri İlker Berktav.

enCRYPTON Labs
Theoretical Cryptography Division

İstanbul, TR
January 27, 2026 – Current

- Exploration of client-efficient fully homomorphic encryption paradigms.

Sabanci University
Cryptography and Information Security Group

İstanbul, TR
August 29, 2024 – Current

- Analysis of algebraic and approximation-theoretic techniques for homomorphic encryption scheme optimization.

Sabanci University
Program for Undergraduate Research (PURE)

İstanbul, TR
Summer 2024, 2025

- Development of accelerators for homomorphic encryption, 2024 Report, 2025 Report.
- Supervisor: Erkey Savaş.
- 2024: Implementation of CKKS bootstrapping and GPU-accelerated algorithms (FFT, Karatsuba).
- 2025: Supervision of two undergraduate students on FHE scheme implementation (TFHE, scheme switching, matrix multiplication).

PUBLICATIONS

- [1] E. İzbudak and K. İ. Berktav, *A derived legendrian category for shifted contact stacks*, 2026. arXiv: 2605.13792 [math.AG].
- [2] E. İzbudak and K. İ. Berktav, *Equivariant quotients of derived symplectic spaces and legendrian intersection theorem*, 2026. arXiv: 2605.08394 [math.AG].
- [3] E. İzbudak, K. İ. Berktav, and M. F. Arıkan, “Derived contact surgery,” Work in progress, 2026.
- [4] E. İzbudak, K. Kaytancı, F. Özbudak, and E. Savaş, “Homomorphic matrix multiplication via self-dual normal bases and twisted polynomial rings,” Work in progress, 2026.
- [5] E. İzbudak, A. Ş. Özcan, and T. Tosun, “Network efficient fully homomorphic encryption,” Work in progress, 2026.

SELECTED TALKS

<i>Contractibility in Set_Δ</i> Notes	Middle East Technical University May 15, 2026
<i>Basics of Set_Δ</i> Notes	Middle East Technical University April 8, 22, 30, 2026
<i>Constructing Spaces</i> Notes	Middle East Technical University December 22, 2026
<i>Moduli Problems and Stacks</i> Notes	Middle East Technical University December 7, 2026
<i>All Morphisms Are Equal, But Some Morphisms Are More Equal Than Others</i> DRP Türkiye, Report, Slides	Sabanci University September 5, 2024
<i>Ciphertext-Ciphertext Matrix Multiplication</i> Slides	Online Winter 2025
<i>HERMES Ring Packing</i> Slides	Online Winter 2025
<i>Cryptanalysis of PLWE Based on Zero-Trace Quadratic Roots</i> Slides	Online Winter 2025
<i>Fast Multiplication and PLWE-RLWE Equivalence</i> Slides	Online Fall 2024
<i>Lattice-Based Foundations of Homomorphic Encryption</i> Slides	Middle East Technical University Fall 2024
<i>Sparse Packing for CKKS</i> Slides	Sabanci University Summer 2024

CONFERENCES & WORKSHOPS

• <i>Higher Categories for the Working Mathematician (Upcoming)</i>	July 27–31, 2026, Göttingen, DE
• <i>Homotopy Theory, K-theory, and Topological Data Analysis (Upcoming)</i>	June 8–12, 2026, London, ON, CA
• <i>Thirty-First Gökova Geometry/Topology Conference (Upcoming)</i>	May 18–23, 2026, Gökova, TR
• <i>Theory of Cryptography Conference</i>	December 1-5, 2025, Aarhus, DK
• <i>Higher Structures Workshop</i>	September 8-12, 2025, Hamburg, DE
• <i>Higher Structures Summer School</i>	September 1-5, 2025, Hamburg, DE
• <i>Random Real Algebraic Geometry Summer School</i>	August 11-15, 2025, İzmir, TR
• <i>Thirtieth Gökova Geometry / Topology Conference</i>	May 25-30, 2025, Muğla, TR
• <i>Mapping Class Groups and Beyond</i>	May 14-18, 2025, Ankara, TR
• <i>Real World Crypto Symposium</i>	March 26-28, 2025, Sofia, BG
• <i>4th Annual FHE.org Conference</i>	March 25, 2025, Sofia, BG
• <i>8th Homomorphic Encryption Standards Meeting</i>	March 23, 2025, İstanbul, TR

LEARNING SEMINARS

Higher Structures

Kadri İlker Berktav, 9 talks contributed.

Middle East Technical University

Spring 2025 – Current

Derived Algebraic Geometry

Alexander Zahrer, Lucy Grossman, Marcus Nicolas, Outline

Online

Spring 2025 – March 19 2026

Recent papers on RLWE/PLWE problems

Ferruh Özbudak, Kamer Kaya, Selim Kırbıyık, 3 talks contributed.

Online

Fall 2024

Serre's Local Algebra

Özgür Kişisel, 2 talks contributed, notes (for my talks).

Middle East Technical University

Spring 2024

Categorical Quantum Mechanics

Cihan Okay

Bilkent University

Spring 2024

CERTIFICATES

- LFD103, The Linux Foundation, February 2023
- LFC108, The Linux Foundation, March 2025

SCHOLARSHIPS

- TMD Cahit Arf Scholar 2024 - 2025
- TOG-Bezmen Merit Scholar 2022 - Current

WORK EXPERIENCE

- Liberior.org, Founder, System Administrator February 2023 – January 2024
Liberior was an organization providing secure and free e-mail/XMPP accounts.
- Liberpub, Founder, System Administrator July 2022 – January 2024
Liberpub was an online printing service with no initial cost for the author.
- Upwork, Freelance Software Developer July 2022 – September 2023
Projects include: $\LaTeX$ to MathJax conversion, PDF harvesting and manipulation.

COMMUNITY CONTRIBUTIONS

- Tor Project, Relay Operator December 2020 – Current
Supporting global internet freedom by maintaining a secure bridge.
- Gentoo, Proxied Maintainer August 2022 – Current
Maintained packages: `sc-im`, `mutt-wizard`, `libxlswriter`
- Qubes, Technical Documentation Writer November 2020 – December 2020
Authored the technical documentation for the `split-ssh` feature.

SKILLS

- Python, Rust, Shell Scripting
- $\LaTeX$, Git, Linux Administration (Gentoo)
- Cryptography (HElib, HEonGPU, Custom CKKS Python Library)
- Digital Forensics (Wireshark, Ghidra, radare2, Volatility)

LANGUAGES

- **English:** Fluent
- **French:** Basic reading
- **Turkish:** Native